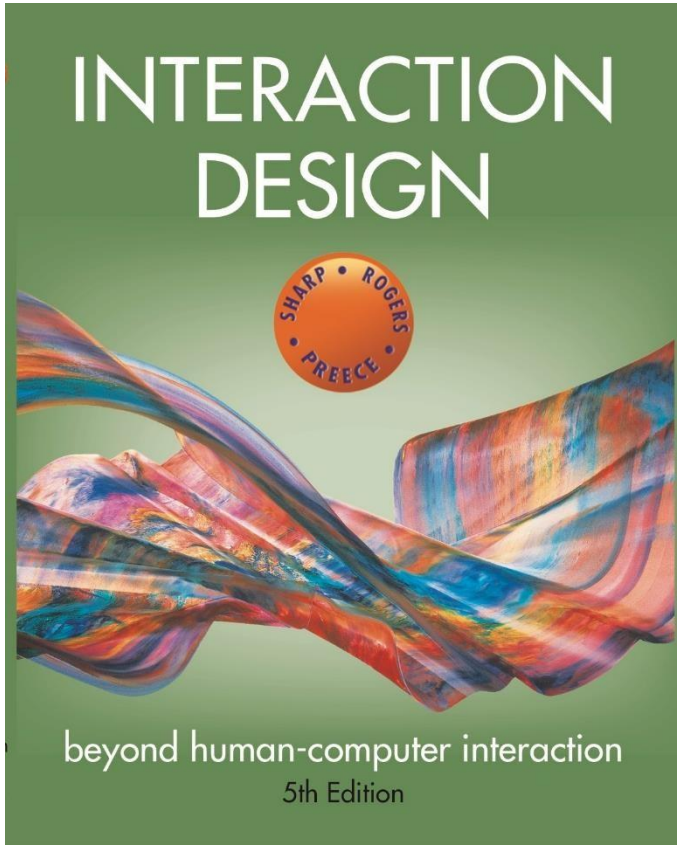

Week 3:
Lecture 5 – Chapter 4
COGNITIVE ASPECTS

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Overview

- What is cognition?
- Why it is important to understand in HCI
- Describe how cognition has been applied to interaction design
- Explain what are mental models and how to elicit them
- Cover relevant theories of cognition

What is cognition?

- Thinking, remembering, learning, daydreaming, decision-making, seeing, reading, talking, writing…
- Ways of classifying cognition at a higher level:
  - Experiential vs. reflective cognition (Norman, 1993)
  - Fast vs slow thinking (Kahneman, 2011)

Fast vs slow thinking?

- $2 + 2 =$
- $21 \times 29 =$

How can understanding cognition help?

- Provides knowledge about **what users can and cannot be expected** to do
- Identifies and explains the **nature and causes of problems that users encounter**
- Provides theories, modeling tools, guidance, and methods that can lead to the design of better interactive products

Cognitive processes

- Attention
- Perception
- Memory
- Learning
- Reading, speaking and listening

Attention

- Selecting things on which to concentrate at a point in time from the mass of stimuli around us
- Allows us to focus on **information that is relevant** to what we are doing
- Involves audio and/or visual senses
- Focused and divided attention
  - Enables us to be selective in terms of the mass of competing stimuli, but limits our ability to keep track of all events
- Design recommendation
  - Information at the interface should be structured to capture users' attention, for example, use perceptual boundaries (windows), color, reverse video, sound, and flashing lights

Activity: Find the price for a double room at the Quality Inn in Pennsylvania

Pennsylvania
Bedford Motel/Hotel: Crinaline Courts
(814) 623-9511 S: $118 D: $120
Bedford Motel/Hotel: Holiday Inn
(814) 623-9006 S: $129 D: $136
Bedford Motel/Hotel: Midway
(814) 623-8107 S: $121 D: $126
Bedford Motel/Hotel: Penn Manor
(814) 623-8177 S: $119 D: $125
Bedford Motel/Hotel: Quality Inn
(814) 623-5189 S: $123 D: $128
Bedford Motel/Hotel: Terrace
(814) 623-5111 S: $122 D: $124
Bradley Motel/Hotel: De Soto
(814) 362-3567 S: $120 D: $124
Bradley Motel/Hotel: Holiday House
(814) 362-4511 S: $122 D: $125
Bradley Motel/Hotel: Holiday Inn
(814) 362-4501 S: $132 D: $140
Breezewood Motel/Hotel: Best Western Plaza
(814) 735-4352 S: $120 D: $127
Breezewood Motel/Hotel: Motel 70
(814) 735-4385 S: $116 D: $118

Activity: Find the price for a double room at the Quality Inn in Pennsylvania

South Carolina

City	Motel/Hotel	Area code	Phone	Rates Single	Rates Double
Charleston	Best Western	803	747-0961	$126	$130
Charleston	Days Inn	803	881-1000	$118	$124
Charleston	Holiday Inn N	803	744-1621	$136	$146
Charleston	Holiday Inn SW	803	556-7100	$133	$147
Charleston	Howard Johnsons	803	524-4148	$131	$136
Charleston	Ramada Inn	803	774-8281	$133	$140
Charleston	Sheraton Inn	803	744-2401	$134	$142
Columbia	Best Western	803	796-9400	$129	$134
Columbia	Carolina Inn	803	799-8200	$142	$148
Columbia	Days Inn	803	736-0000	$123	$127
Columbia	Holiday Inn NW	803	794-9440	$132	$139
Columbia	Howard Johnsons	803	772-7200	$125	$127
Columbia	Quality Inn	803	772-0270	$134	$141
Columbia	Ramada Inn	803	796-2700	$136	$144
Columbia	Vagabond Inn	803	796-6240	$127	$130

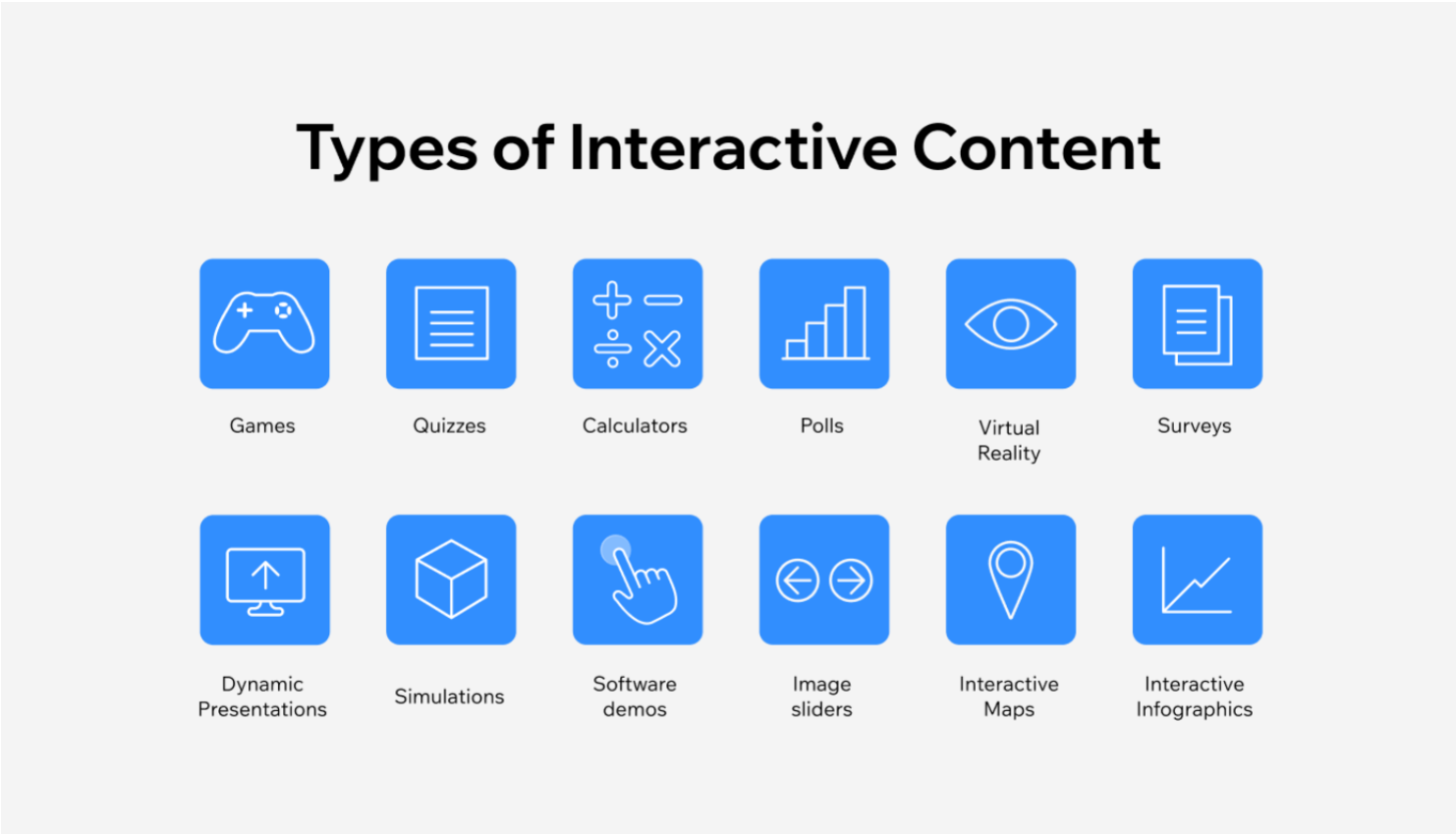
Activity - Results

- Tullis (1987) found that the two screens produced quite different results
  - 1st screen: Took an average of 5.5 seconds to search
  - 2nd screen: Took 3.2 seconds to search
- Why? Since both displays have the same density of information (31percent)?
- Spacing
  - In the 1st screen, the information is bunched up together, making it hard to search
  - In the 2nd screen, the characters are grouped into vertical categories of information making it easier

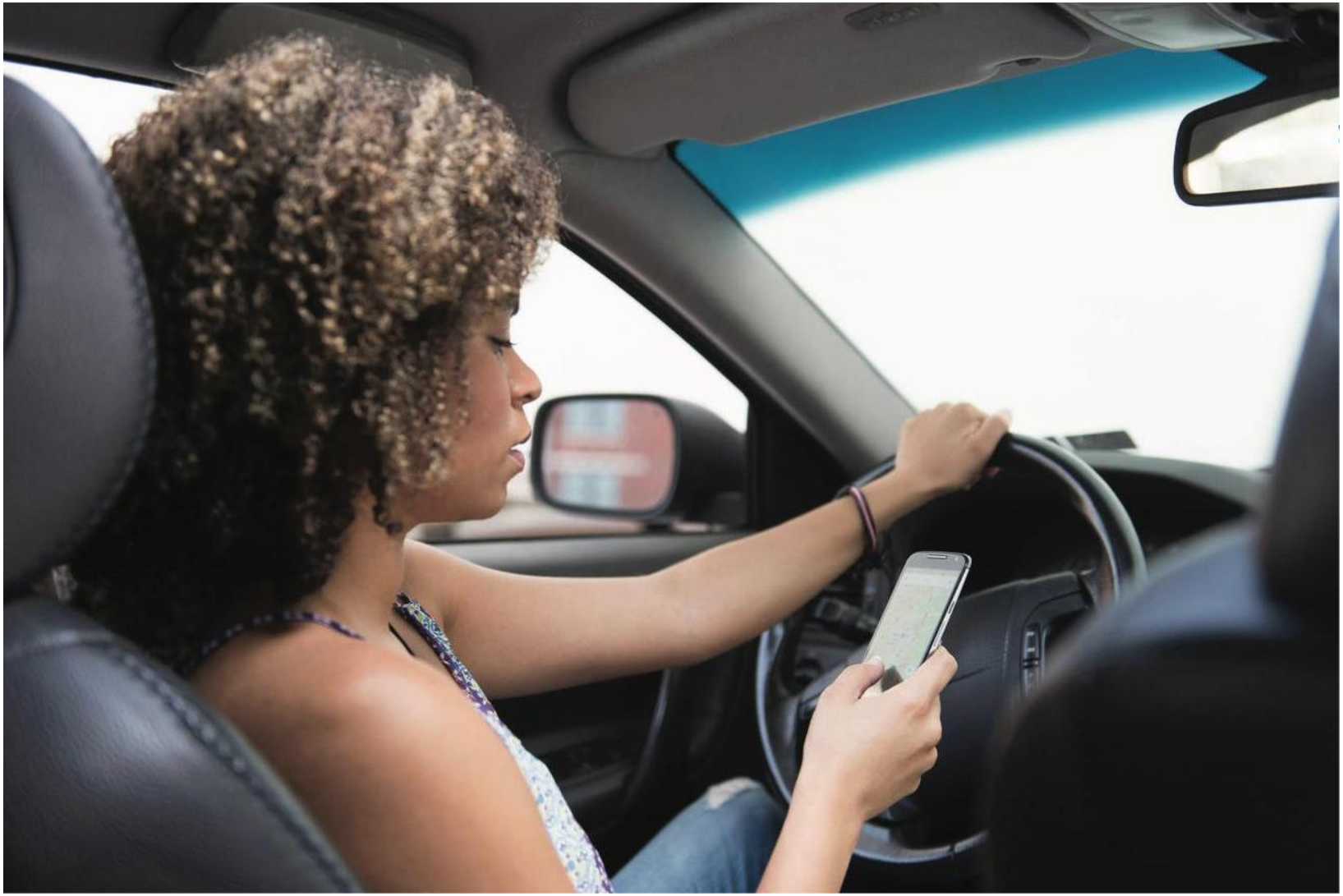
Multitasking and Attention

- It is increasingly common for workers to multitask
- For example, hospital workers have to attend to multiple screens in an operating room that provide new kinds of real-time information
- This requires clinician's constant attention to check if any data is unusual or anomalous
- Need to develop new attention and scanning strategies

Attention and scanning strategies



Is it okay to use a phone when driving?



No!

- Driving is very demanding
- Drivers are prone to being distracted
- There is a significant chance of causing accidents
- Drivers' reaction times are longer to external events when talking on the phone in a car (Caird et al., 2018)
- Drivers using their phones rely more on their expectations about what is likely to happen next as conducting a conversation takes up their attention

Are hands-free phones safer to use when driving?

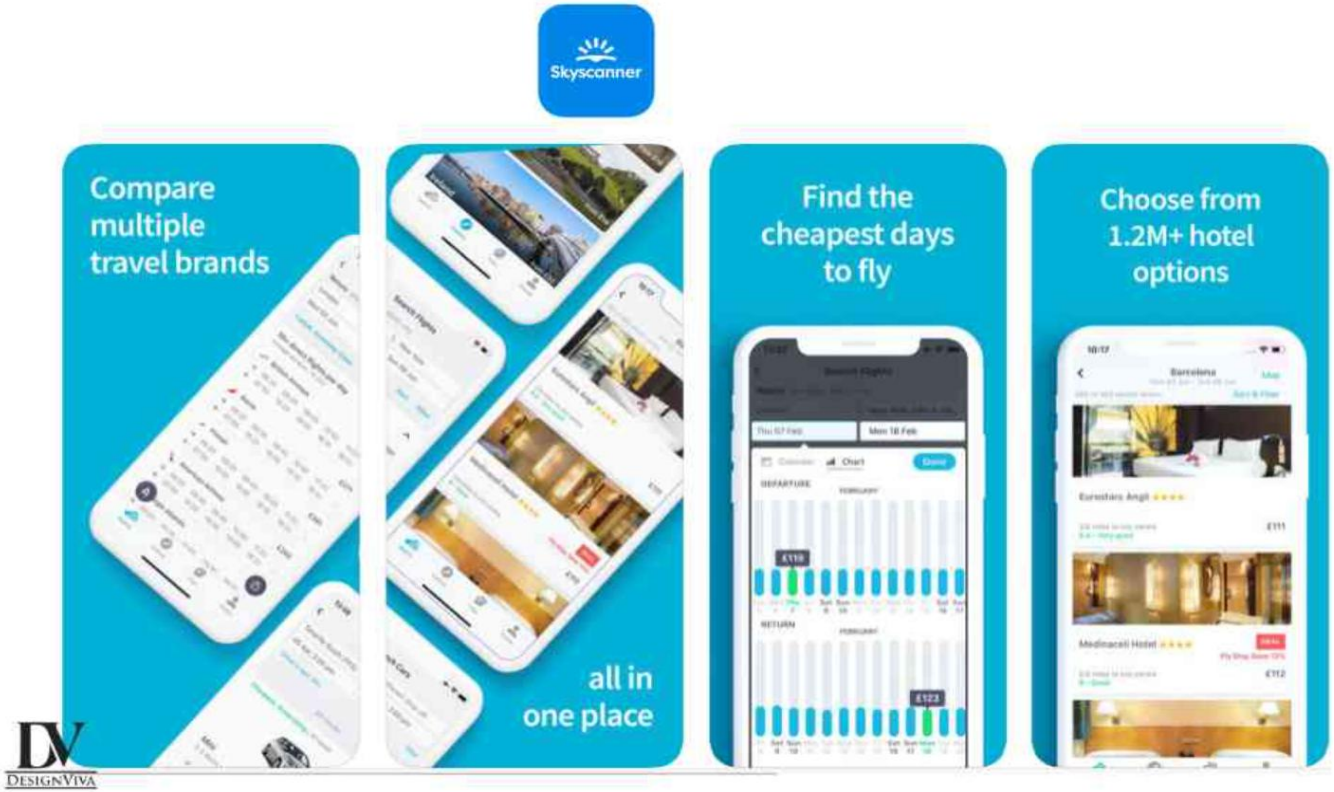
- No, as same type of cognitive processing is happening when talking

- The same thing happens when talking with front seat passenger
  - But both can stop in mid-sentence if a hazard is spotted allowing the driver to switch immediately to the road
  - So, it's less dangerous talking to a front seat passenger than a remote person
  - A remote person on the end of a phone is not privy to what the driver is seeing and will carry on the conversation when there is a hazard
  - This makes it difficult for the driver to switch all their attention to the road

Design implications for attention

- Context: Make information salient when it needs to be attended to at a given stage of a task
- Use techniques to achieve this:
  - **For example, color, ordering, spacing, underlining, sequencing, and animation**
- Avoid cluttering visual interfaces with **too much information**
- Consider designing different ways to support effective **switching and returning** to an interface

Skyscanner: An example of a good design



Skyscanner's app design focuses on simplicity and efficiency. With a user-friendly interface, the app allows travelers to search for flights, hotels, and car rentals effortlessly. The "Everywhere" feature, which suggests destinations based on budget and preferences, adds an element of spontaneity to trip planning.

Perception

- How information is acquired from the world and transformed into experiences

- Obvious implication is to design representations that are readily perceivable, for instance:
  - Text should be legible
  - Icons should be easy to distinguish and read

Is color contrast good? Find Italian

Black Hills Forest	Peters Landing	Jefferson Farms	Devlin Hall
Cheyenne River	Public Health	Psychophysics	Positions
Social Science	San Bernardino	Political Science	Hubard Hall
South San Jose	Moreno Valley	Game Schedule	Fernadino Beach
Badlands Park	Altamonte Springs	South Addision	Council Bluffs
Juvenile Justice	Peach Tree City	Cherry Hills Village	Classical Lit
Results and Stats	Highland Park	Creative Writing	Sociology
Thousand Oaks	Machesney Park	Lake Havasu City	Greek
Promotions	Vallecito Mts.	Engineering Bldg	Wallace Hall
North Palermo	Rock Falls	Sports Studies	Concert Tickets
Credit Union	Freeport	Lakewood Village	Public Radio FM
Wilner Hall	Slaughter Beach	Rock Island	Children's Museum
Performing Arts	Rocky Mountains	Deerfield Beach	Writing Center
Italian	Latin	Arlington Hill	Theater Auditions
Coaches	Pleasant Hills	Preview Game	Delaware City
Mckees Rocks	Observatory	Richland Hills	Scholarships
Glenwood Springs	Public Affairs	Experts Guids	Hendricksville
Urban Affairs	Heskett Center	Neff Hall	Knights Landing
McLeansboro	Brunswick	Grand Wash Cliffs	Modern Literature
Experimental Links	East Millinocket	Indian Well Valley	Studio Arts
Graduation	Women's Studies	Online Courses	Hugher Complex
Emory Lindquist	Vacant	Lindquist Hall	Cumberland Flats
Clinton Hall	News Theatre	Fisk Hall	Central Village
San Luis Obispo	Candlewood Isle	Los Padres Forest	Hoffman Estates

Are borders and white space better? Find French

Webmaster	Curriculum	Student Life	Dance
Russian	Emergency (EMS)	Accountancy	Gerontology
Athletics	Statistics	McKnight Center	Marketing
Go Shockers	Award Documents	Council of Women	College Bylaws
Degree Options	Language Center	Commute	Why Wichita?
Newsletter	Future Shockers	Small Business	Tickets
Geology	Intercollegiate	Thinker & Movers	Career Services
Manufacturing	Bowling	Alumni	Doers & Shockers
Management	Wichita Gateway	Foundations	Core Values
UCATS	Transfer Day	Corbin Center	Grace Wilkie Hall
Alumni News	Job Openings	Jardine Hall	Strategic Plan
Saso	Live Radio	Hugo Wall School	Medical Tech
Educational Map	Beta Alpha Psi	Staff	Softball, Men's
Physical Plant	Liberal Arts	Aerospace	McKinley Hall
Graphic Design	Counseling	Choral Dept.	Email
Non Credit Class	Biological Science	Alberg Hall	Dental Hygiene
Media Relations	Duerksen Fine Art	French	Tenure
Advertising	EMT Program	Spanish	Personnel Policies
English	Religion	Parents	Instrumental
Graduate Complex	Art Composition	Wrestling	Nursing
Music Education	Physics	Philosophy	Opera
Advising Center	Entrepreneurship	Wichita Lyceum	Sports History
Medical School	Koch Arena	Fairmount Center	Athletic Dept.
Levitt Arena	Roster	Women's Museum	Health Plan

Activity

- Weller (2004) found people took less time to locate items for information that was grouped
  - Using a border (2nd screen) compared with using color contrast (1st screen)
- Some argue that too much white space on web pages is detrimental to search process
  - Makes it hard to find information
- Do you agree?

Activity: Which is the easiest to read and why?

What is the time?

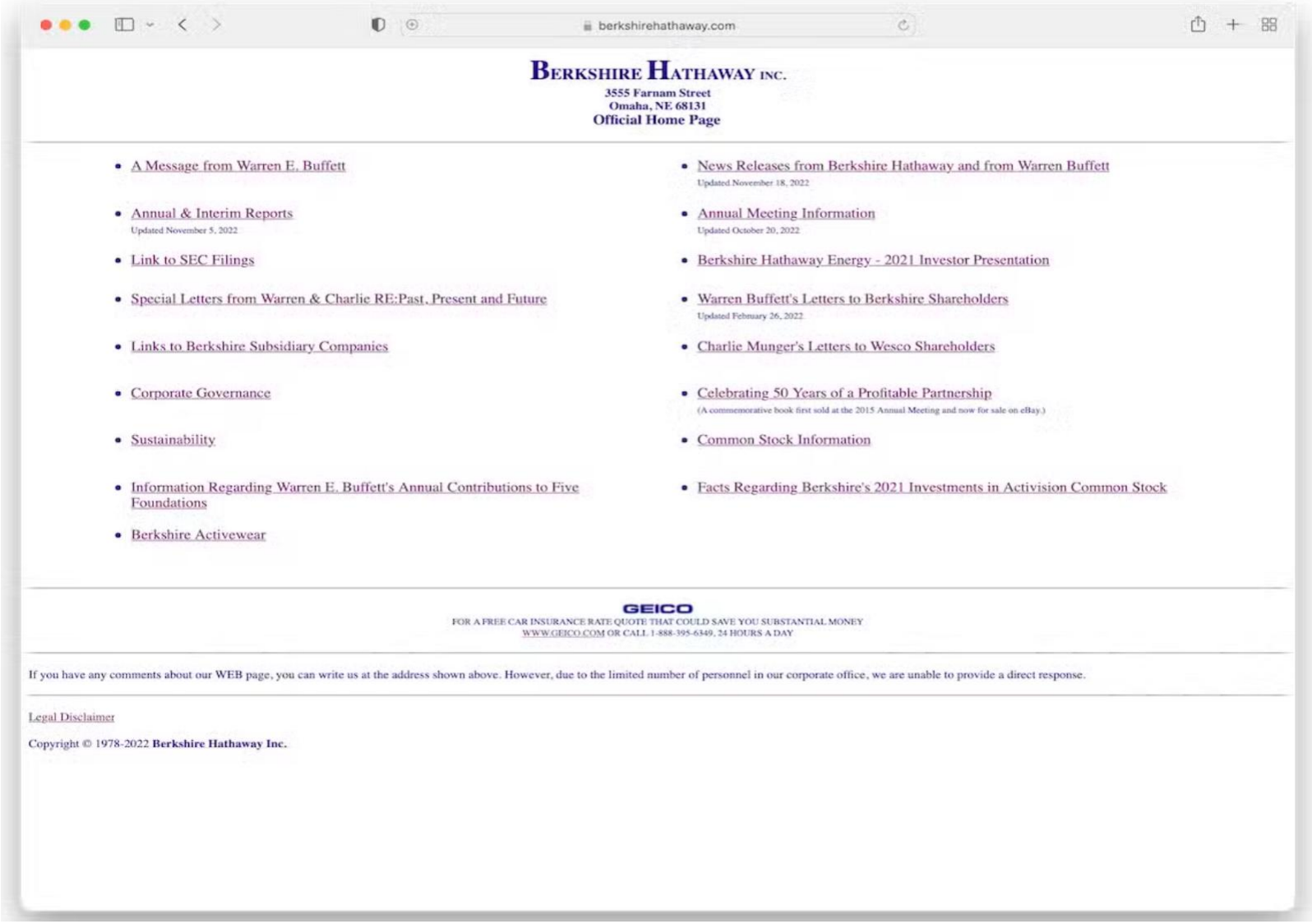
What is the time?

What is the time?

What is the time?

What is the time?

A bad website design: Too much white



https://colorlib.com/wp/bad-websites/

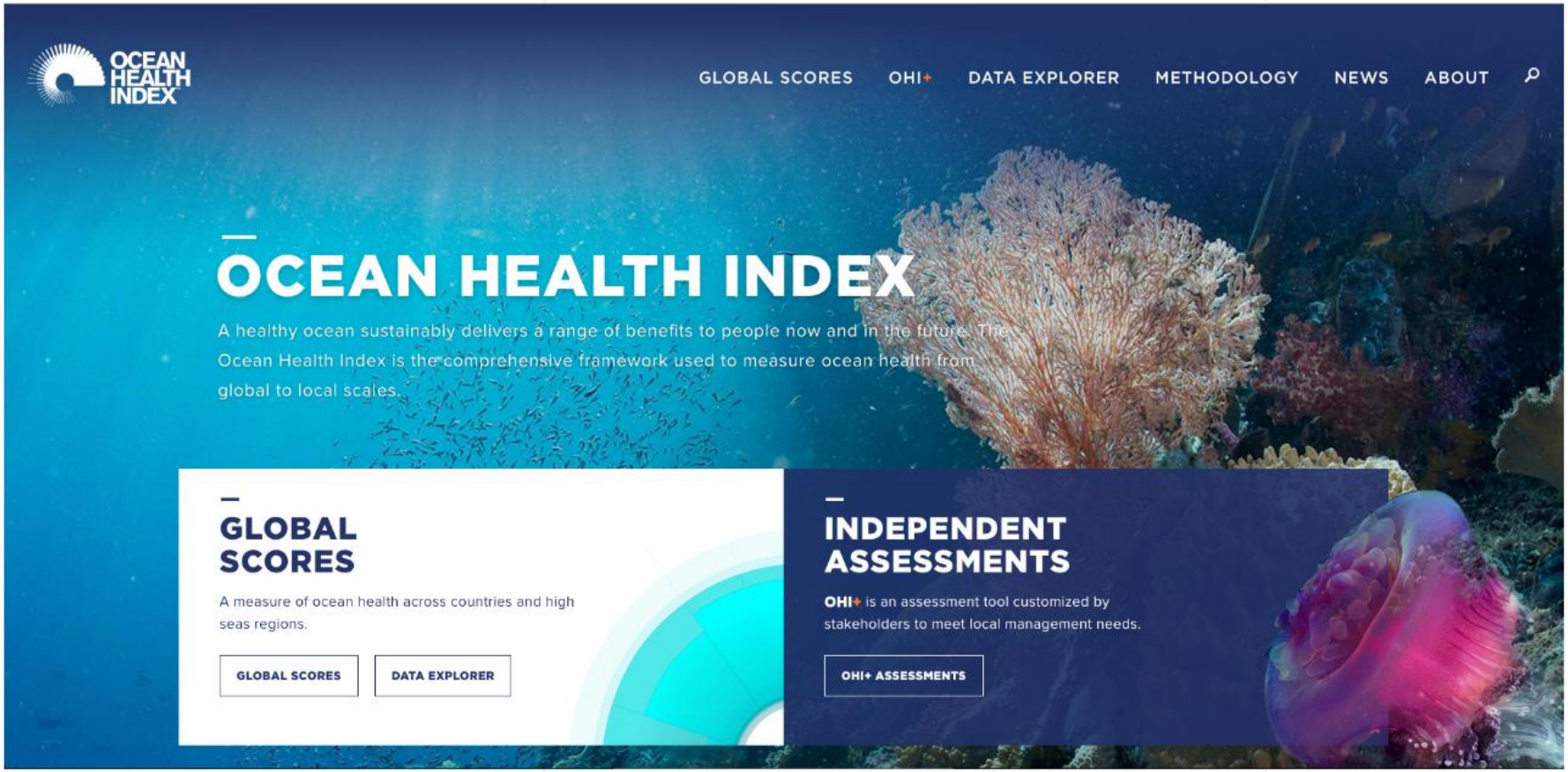
A bad website design: Too much info

https://colorlib.com/wp/bad-websites/

Design implications

- Icons should enable users to *distinguish* their meaning readily
- Bordering and spacing are effective visual ways of grouping information
- Sounds should be audible and distinguishable
- Research proper color contrast techniques when designing an interface:
  - Yellow on black or blue is fine
  - Yellow on green or white is a no-no
- Haptic feedback should be used judiciously

A good web design



Memory

- Involves **recalling** various kinds of knowledge that allow people to act appropriately
  - For example, recognizing someone's face or remembering someone's name

- We don't remember everything—it involves **filtering and processing** what is attended to

Memory

- **Context is important** as to how we remember (that is, where, when, how, and so on)

- We **recognize** things much better than being able to **recall** things

- We remember less about objects that we have **photographed** than when we observe them with the **naked eye** (Henkel, 2014)

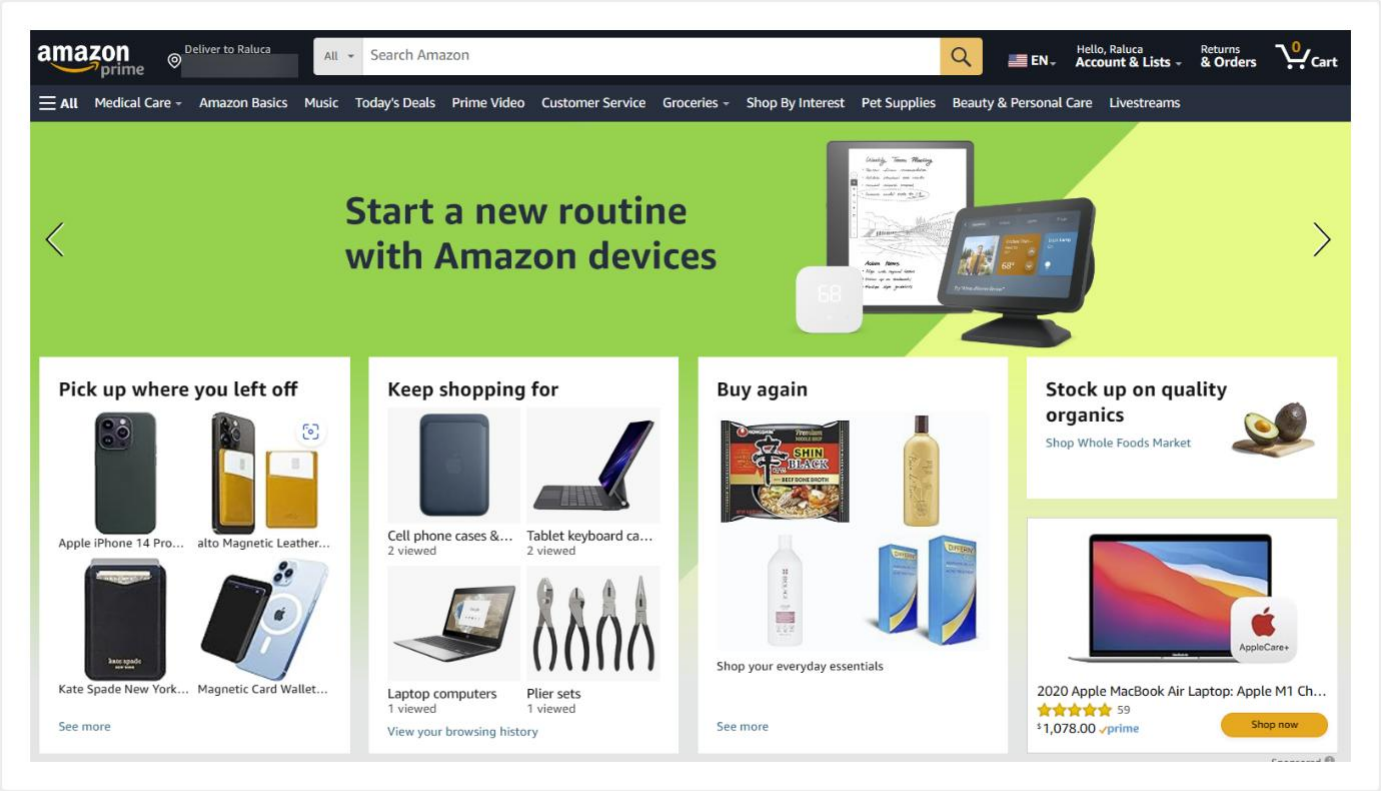
Processing in memory

- Encoding is first stage of memory
  - Determines which information is attended to in the environment and how it is interpreted
- The more attention paid to something…
- The more it is processed in terms of thinking about it and comparing it with other knowledge…
- The more likely it is to be remembered
  - For example, when learning about HCI, it is better to reflect upon it, carry out exercises, have discussions about it, and write notes than just passively read a book, listen to a lecture or watch a video about it

Activity

- Try to remember the dates of your grandparents' birthday
- Try to remember the cover of the last two books you read
- Which was easiest? Why?
- People are very good at remembering visual cues about things
  - For instance, the color of items, the location of objects and marks on an object
- They find it more difficult to learn and remember arbitrary material
  - For example, birthdays and phone numbers

Amazon’s web design



When a user goes back to Amazon.com, the personalized homepage includes a list of recently viewed and purchased items.

https://www.nngroup.com/articles/recognition-and-recall/

Recognition versus recall

- Command-based interfaces require users to recall from memory a name from a possible set of 100s of names
- Graphical interfaces provide visually-based options (menus, icons) that users need only browse through until they recognize one
- Web browsers provide tabs and history lists of visited URLs that support recognition memory

The problem with the classic '7,+ or −2'

- George Miller's (1956) theory of how much information people can remember
- People's immediate memory capacity is very limited to 7, + or −2
- Has been applied in interaction design when considering how many options to display
- But is it a good use of a theory in HCI?
- Is it helpful?

When creating an interface, should the designer…

- Present only 7 options on a menu
- Display only 7 icons on a tool bar
- Have no more than 7 bullets in a list
- Place only 7 items on a pull down menu
- Place only 7 tabs on the top of a website page?
- Not necessarily…

The reason is…

- People can scan lists of bullets, tabs, and menu items for the one they want
- They don't have to recall them from memory, having only briefly heard or seen them
- So you can have more than seven at the interface
  - For instance, history lists of websites visited
- Sometimes a small number of items is good
  - For example, smart watch displays
- Depends on task and available screen estate

Personal Information management

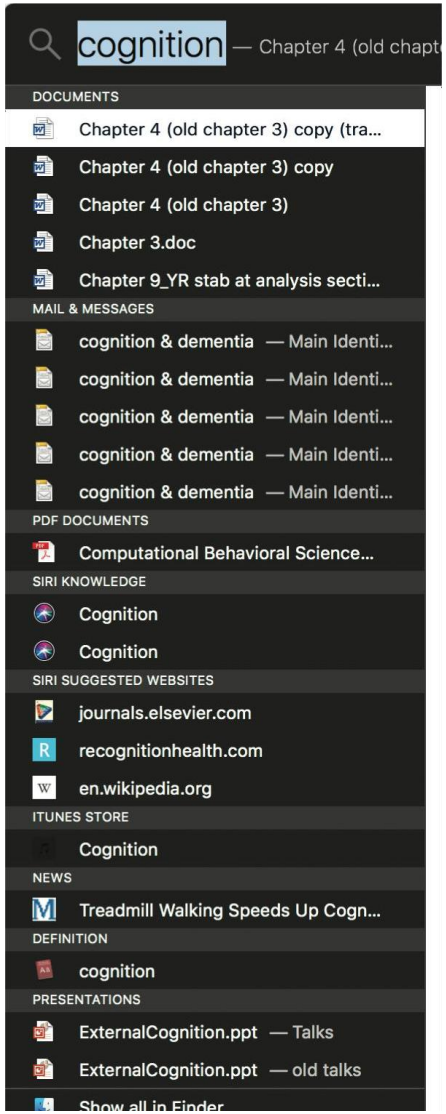
Is a growing problem for many users:

- They accumulate a vast numbers of documents, images, music files, video clips, emails, attachments, bookmarks, and so forth
- Where and how to save them all; then remembering what they were called and where to find them again
- Naming most common means of encoding them
- But can be difficult to remember, especially when you have 10,000s
- How might such a process be facilitated taking into account people's memory abilities?

Personal Information management

- Most common approach is to use folders and naming
- Strong preference for scanning across and within folders when looking for something
- Search engines only helpful if you know the name of the file
- Smart search engines help with listing relevant files for partial name or when type in first letter

Apple's Spotlight search tool



Memory load

- Online/mobile and phone banking now require users to provide multiple pieces of information to access their account
  - For instance, ZIP code, birthplace, a memorable date, first school attended
  - Known as multifactor authentication (MFA)
- Why?
  - Increased security concerns
- Password managers, such as LastPass, have been developed that require only one master password
  - Reduces stress and memory load on users
- Passwords could become extinct with the widespread use of biometrics and computer vision algorithms
- https://www.lastpass.com/

Design implications

- Reduce cognitive load by avoiding **long and complicated procedures** for carrying out tasks
- Design interfaces that promote **recognition rather than recall**
- Provide users with **various ways of labelling digital information** to help them easily identify it again
  - For example, folders, categories, color, flagging, and time stamping

Learning

- Involves the **accumulation of skills and knowledge** involving memory
- Two main types:
  - Incidental learning
    - (for example, recognizing people's faces, what you did today)
  - Intentional learning
    - (for instance, studying for an exam, learning to cook)
  - Intentional learning is much harder!
  - Many technologies have been developed to help (for example, multimedia, animations, VR)
- People find it hard to learn by following instructions in a manual
- People prefer to learn by doing

Design implications

- Design interfaces that encourage **exploration**
- Design interfaces that constrain and **guide learners**
- **Dynamically linking** concepts and representations can facilitate the learning of complex material

Reading, speaking, and listening

The ease with which people can read, listen, or speak differs:

- Many prefer listening to reading
- Reading can be quicker than speaking or listening
- Listening requires less cognitive effort than reading or speaking
- Dyslexics have difficulties understanding and recognizing written words

Applications

- Voice user interfaces allow users to interact with them by asking questions
  - For example, Google Voice, Siri, and Alexa
- Speech-output systems use artificially-generated speech
  - For instance, written text-to-speech systems for the visually impaired
- Natural-language systems enable users to type in questions and give text-based responses
  - Such as, chatbots

Design implications

- Speech-based menus and instructions should be **short**
- **Accentuate** the intonation of artificially generated speech voices
  - They are harder to understand than human voices
- Provide opportunities for making **text large** on a screen